

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

HOPE MEDICAL ENTERPRISES, INC.,
Petitioner,

v.

OREGON HEALTH & SCIENCE UNIVERSITY,
Patent Owner.

IPR2022-00123
Patent 10,596,190 B2

Before ZHENYU YANG, ROBERT A. POLLOCK, and DAVID COTTA,
Administrative Patent Judges.

YANG, *Administrative Patent Judge.*

JUDGMENT

Final Written Decision

Determining Challenged Claim 1 Unpatentable
Granting in Part and Denying in Part Patent Owner's
Revised Motion to Amend
35 U.S.C. § 318(a)

I. INTRODUCTION

Hope Medical Enterprises, Inc. (“Petitioner”) filed a Petition (Paper 1, “Pet.”), seeking an *inter partes* review of the single claim of U.S. Patent No. 10,596,190 B2 (Ex. 1001, “the ’190 patent”). We instituted trial to review claim 1. Paper 10.

After institution, Oregon Health & Science University (“Patent Owner”) did not file a Response to the Petition. Instead, it filed a Non-Contingent Motion to Amend, requesting that we cancel claim 1 and replace it with proposed substitute claim 2. Paper 16 (“MTA”), 1. Petitioner filed an Opposition to the MTA. Paper 19 (“MTA Opp.”). We issued a Preliminary Guidance on the MTA, indicating our initial, preliminary, non-binding views that Petitioner had established a reasonable likelihood that the proposed substitute claim was unpatentable. Paper 20 (“PG”).

Thereafter, Patent Owner filed a Revised Non-Contingent Motion to Amend (Paper 21, “Revised MTA”), Petitioner filed an Opposition to the Revised Motion to Amend (Paper 26, “Revised MTA Opp.”), Patent Owner filed a Reply (Paper 28), and Petitioner filed a Sur-reply (Paper 31).

An oral hearing for this proceeding was held on March 7, 2023, and the transcript of that hearing is of record. *See* Paper 34 (“Tr.”).

The Board has jurisdiction under 35 U.S.C. § 6 and issues this Final Written Decision pursuant to 35 U.S.C. § 318 and 37 C.F.R. § 42.73. For the reasons provided below, we grant the Revised MTA as to cancelling original claim 1 of the ’190 patent, but deny it as to adding proposed substitute claim 2.

A. Related Matters

According to the parties, the '190 patent is not the subject of any district court litigation. Pet. 77; Paper 5, 1.

On January 11, 2022, Patent Owner filed a Request for Supplemental Examination of the '190 patent, Application No. 96/000,390. Paper 6, 1; Ex. 2003. On March 9, 2022, the Office ordered an *ex parte* reexamination of claim 1 of the '190 patent pursuant to 35 U.S.C. § 257. Ex. 1035. On the day we instituted trial, we issued an Order staying the *ex parte* reexamination pending completion of this *inter partes* review. Paper 11.

B. The '190 Patent

The '190 patent relates to a method for eliminating or reducing ototoxicity in pediatric patients receiving a platinum-based chemotherapeutic. Ex. 1001, Abstract; 1:22–25.

Platinum-based therapeutics are important in treating a variety of pediatric malignancies. *Id.* at 1:29–31. Unfortunately, at commonly used doses and schedules, platinum-based therapeutics, such as cisplatin and carboplatin, often cause hearing loss. *Id.* at 1:33–39. According to the '190 patent, there was a need for “safe and effective methods for treating pediatric patients to reduce ototoxicity and hearing loss in these patients that do not compromise the efficacy of the platinum-based therapeutic.” *Id.* at 1:49–52.

According to the '190 patent, “STS [(sodium thiosulfate)] is a water-soluble thiol compound with reducing agent properties and is commercially available as an established antidote for acute cyanide poisoning.” *Id.* at 8:61–63. STS “has been used in oncology for preventing

cisplatin nephrotoxicity, carboplatin ototoxicity and as an antidote for extravasation of various chemotherapy agents.” *Id.* at 8:63–67. The ’190 patent describes administering an effective amount of STS to a pediatric patient following the administration of a platinum-based chemotherapeutic. *Id.* at 1:59–62. According to the ’190 patent, “the administration of STS was found to not adversely affect the efficacy of the platinum based chemotherapeutic and decreased the incidence and severity of ototoxicity in pediatric patients.” *Id.* at 1:62–65.

II. CANCELLED CLAIM 1

Under our Rules, a motion to amend may cancel a challenged claim. 37 C.F.R. § 42.121(a)(3). Patent Owner requests that we non-contingently cancel claim 1 and enter proposed substitute claim 2. MTA 1. Patent Owner’s request to cancel claim 1 is granted.

III. ANALYSIS

A. *Proposed Substitute Claim 2*

In the Revised MTA, Patent Owner proposes substitute claim 2 to replace claim 1. Proposed substitute claim 2, with additions to claim 1 shown in underline and deletions shown in double brackets, is reproduced below:

2. A method of reducing ototoxicity in a human pediatric patient of [[about]] less than 5 years of age [[and under]], comprising: administering:

(I) cisplatin to treat a localized and non-metastatic medulloblastoma tumor at a dose specified by a pediatric medulloblastoma treatment plan [[of about 1 mg/kg to about 5 mg/kg or about 10 mg/m² to about 300 mg/m² per cycle, administered on between about 1 and 5 days per cycle]]; and

(II) sodium thiosulfate, at a dose of $[[\text{about } 5 \text{ g/m}^2 \text{ to about } 25 \text{ g/m}^2]]$ about 15 g/m^2 given after each dose of the cisplatin $[[\text{on between about } 1 \text{ and } 5 \text{ days per cycle}]]$, about six hours after the completion of the administration of each dose of the cisplatin,

wherein the reduction of ototoxicity is at least about $[[50\%]]$ 60% better than a patient not receiving an administration of sodium thiosulfate;

wherein ototoxicity is determined by one or more criteria selected from:

- a) a reduction in hearing measured by a 20 dB loss at a single frequency;
- b) a reduction in hearing measured by a 10 dB loss at two consecutive frequencies;
- c) loss of response in three consecutive test frequencies where responses were previously obtained;
- d) a reduction in bilateral high-frequency hearing characterized by the criteria;

$[[\text{(i)}]]$ (i) a 40 dB hearing loss at all frequencies, which indicates a grade O or minimal hearing loss;

$[[\text{(ii)}]]$ (i) a ≥ 40 dB hearing loss at 8,000 Hz only, which indicates a grade 1 or mild hearing loss;

$[[\text{(iii)}]]$ (ii) a ≥ 40 dB hearing loss at 4,000 Hz and above, which indicates a grade 2 or moderate hearing loss;

$[[\text{(iv)}]]$ (iii) a ≥ 40 dB hearing loss at 2,000 Hz and above, which indicates a grade 3 or marked hearing loss;

$[[\text{(v)}]]$ (iv) a ≥ 40 dB hearing loss at 1,000 Hz and above, which indicates a grade 4 or severe hearing loss; or

e) a reduction in hearing characterized by the criteria:

$[[\text{(i)}]]$ (i) a ≤ 20 dB hearing loss at all frequencies, which indicates a grade 0 hearing loss;

[[(ii)] (i) a >20 dB HL above 4,000 Hz, which indicates a grade 1 hearing loss;

[[(iii)] (ii) a >20 dB HL at 4,000 Hz and above, which indicates a grade 2 hearing loss;

[[(iv)] (iii) a >20 dB HL at 2,000 Hz or 3,000 Hz, which indicates a grade 3 hearing loss;

[[(v)] (iv) a >40 dB HL at 2,000 Hz and above, which indicates a grade [[1]] 4 hearing loss;

wherein a change in hearing is computed relative to baseline measures prior to the patient receiving cisplatin [[a platinum based chemotherapeutic or sodium thiosulfate or both]]; and

wherein the administration of sodium thiosulfate does not substantively affect relapse free survival or overall survival in the medulloblastoma pediatric patient compared to a medulloblastoma pediatric patient administered cisplatin but not administered sodium thiosulfate.

Relevant to our Decision are the following two limitations:

(1) the ototoxicity efficacy limitation: “wherein the reduction of ototoxicity is at least about 60% better than a patient not receiving an administration of sodium thiosulfate;” and

(2) the overall survival efficacy limitation: “wherein the administration of sodium thiosulfate does not substantively affect . . . overall survival in the medulloblastoma pediatric patient compared to a medulloblastoma pediatric patient administered cisplatin but not administered sodium thiosulfate.”

When analyzing these two limitations, we keep in mind that both have the built-in requirements for the patient—“less than 5 years of age” and with a “localized and non-metastatic medulloblastoma tumor.”

B. Asserted Grounds of Unpatentability

Petitioner challenges the patentability of claim 2 on the following grounds:

Claim Challenged	35 U.S.C. §	References/Basis
2	103	Freyer Presentation, ¹ Adherex Press Release, ² Doolittle, ³ Brock 2016 ⁴
2	112(a)	Enablement

In support of their respective arguments, Petitioner relies on the Declaration of Robert Earhart, M.D., Ph.D. (Ex. 1007), and Patent Owner relies on the Declaration of Shahrard R. Rassekh, M.D., M.H.Sc (Ex. 2024).

C. Principles of Law Concerning a Motion to Amend

In an *inter partes* review, amended claims are not added to a patent as of right, but rather must be proposed as a part of a motion to amend.

¹ACCL0431: A Randomized Phase III Study of Sodium Thiosulfate (STS) for Prevention of Cisplatin-induced Hearing Loss in Children, Presented by David Freyer at 2014 ASCO Annual Meeting (“Freyer Presentation,” Ex. 1003).

²NewsRoom, Adherex Announces Results from Two Sodium Thiosulfate Phase 3 Studies Presented at the 50th American Society of Clinical Oncology (ASCO) Meeting, 1–4 (June 2, 2014) (“Adherex Press Release,” Ex. 1004).

³ Doolittle et al., *Delayed Sodium Thiosulfate as An Otoprotectant Against Carboplatin-Induced Hearing Loss in Patients with Malignant Brain Tumors*, 7 CLIN. CANCER RES. 493–500 (2001) (Ex. 1051).

⁴ Brock et al., Two-year results of clinical efficacy of cisplatin in combination with sodium thiosulfate (STS) vs cisplatin alone in a randomized phase III trial for standard risk hepatoblastoma (SR-HB): SIOPEL 6, 34 J. CLIN. ONCOL., Issue 15, Suppl., Abstract 10514 (2016) (Ex. 1058).

35 U.S.C. § 316(d). Ordinarily, the petitioner “bears the burden of persuasion to show, by a preponderance of the evidence, that any proposed substitute claims are unpatentable.” 37 C.F.R. § 42.121(d)(2); *Lectrosonics, Inc. v. Zaxcom, Inc.*, IPR2018-01129, Paper 15 (“*Lectrosonics*”), 3–4 (PTAB Feb. 25, 2019) (precedential); *Bosch Auto. Serv. Sols. LLC v. Iancu*, 878 F.3d 1027, 1040 (Fed. Cir. 2017).

But before considering the patentability of the proposed substitute claim, we first must determine whether the Revised MTA meets the statutory and regulatory requirements set forth in 35 U.S.C. § 316(d) and 37 C.F.R. § 42.121. *Lectrosonics* at 4. Patent Owner bears the burden of persuasion to show that: (1) the amendment proposes a reasonable number of substitute claims; (2) the amendment responds to a ground of unpatentability involved in the trial; (3) the amendment does not seek to enlarge the scope of the claims of the patent or introduce new subject matter; and (4) the original disclosure sets forth written description support for each proposed claim. *Id.* at 4–8.

Specifically, an amendment may not introduce new matter. 35 U.S.C. § 316(d)(3); 37 C.F.R. § 42.121(a)(2)(ii). New subject matter is any addition to the claims that lacks sufficient support in the subject patent’s original disclosure. *See TurboCare Div. of Demag Delaval Turbomach. Corp. v. Gen. Elec. Co.*, 264 F.3d 1111, 1118 (Fed. Cir. 2001) (“When [an] applicant adds a claim . . . , the new claim[] . . . must find support in the original specification.”). Thus, a motion to amend must set forth: “(1) The support in the original disclosure of the patent for each claim that is added or amended; and (2) The support in an earlier-filed disclosure for each claim for which

benefit of the filing date of the earlier filed disclosure is sought.”

37 C.F.R. §§ 42.121(b)(1), 42.121(b)(2).

The test for new matter is whether the original application’s disclosure “reasonably conveys to those skilled in the art that the inventor had possession of the claimed subject matter as of the filing date.” *Ariad Pharms., Inc. v. Eli Lilly & Co.*, 598 F.3d 1336, 1351 (Fed. Cir. 2010) (en banc). It “requires an objective inquiry into the four corners of the specification from the perspective of a person of ordinary skill in the art” to determine whether the specification “describe[s] an invention understandable to that skilled artisan and show that the inventor actually invented the invention claimed.” *Id.*

D. Level of Ordinary Skill in the Art

Petitioner contends that an ordinarily skilled artisan “would have possessed an advanced degree, such as a Ph.D., M.D., Pharm.D., and two years of experience with the treatment of cancer, including platinum-based therapy such as cisplatin.” Pet. 7–8 (citing Ex. 1007 ¶¶ 26–29).

Patent Owner argues that Petitioner’s proposed definition of the level of ordinary skill is “overly broad.” MTA 7. According to Patent Owner, an ordinarily skilled artisan

would have at least the following qualifications: (a) an M.D. or equivalent medical degree, (b) residency or equivalent training in pediatrics, (c) completion of a fellowship or equivalent training program in pediatric oncology, and (d) at least 2 years in treating central nervous system cancers such as medulloblastoma with platinum-based therapies.

Id.; Revised MTA 11 (citing Ex. 2024 ¶¶ 73–90). Patent Owner appears to challenge the qualification of Dr. Earhart, Petitioner’s declarant, as an ordinarily skilled artisan. MTA 9.

Petitioner contends Patent Owner’s proposed definition of the level of ordinary skill is inappropriate because it “requires qualifications that exclude even the sole inventor of the ’190 Patent, Dr. Edward Neuwelt.”

MTA Opp. 1. According to Petitioner, Dr. Neuwelt is “a neurosurgeon,” not a pediatric oncologist, and lacks the requisite “at least 2 years” experience in platinum-based therapies. *Id.* (citing Ex. 1003, 18; Ex. 1038, 98:9–17, 100:15–101, 4). We agree with Petitioner.

It is well settled that inventors are typically persons skilled in the field of the invention. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1313 (Fed. Cir. 2005) (en banc). Petitioner presents persuasive evidence to show that under Patent Owner’s proposed definition of the level of ordinary skill, Dr. Neuwelt, the sole inventor, would not qualify as an ordinarily skilled artisan, because he is not a pediatric oncologist, even though he is considered “one of the world leading experts on this compound [STS] and otoprotection” and “an expert in the area of [STS] as a . . . drug to use in children with medulloblastoma.” Ex. 1038, 98:9–17, 99:17–18, 101:18–102:13.

Citing Dr. Earhart’s testimony, Petitioner argues “a POSITA ‘may work as part of a multidisciplinary team and draw not only upon his or her own skills, but also take advantage of specialized skills of other team members.’” MTA Opp. 2 (quoting Ex. 1007 ¶ 26). The testimony of Dr. Rassekh, Patent Owner’s expert, supports this argument. Indeed,

Dr. Rassekh testified that “there [are] a number of people on the team that are involved in . . . administering a drug to a patient.” Ex. 1038, 80:19–22; *see also id.* at 99:7–20 (testifying both a researcher with expertise on STS and a pediatric oncologist are needed to “design a study together” to look the subject matter of the ’190 patent); 131:19–132:7 (testifying that a study committee, including audiologist, pharmacologist, pediatric oncologist, and statistician, would draft the study protocol).

Petitioner contends “Dr. Earhart – a pharmacologist and oncologist with decades of experience in both treating patients with cisplatin, and developing clinical protocols – more than qualifies as a POSITA.”

MTA Opp. 2. We agree. For example, Dr. Earhart, an M.D. and a Ph.D., has “worked in academia and in industry as a physician and clinical pharmacologist since 1976.” Ex. 1007 ¶¶ 5, 6. He “first began treating patients [with advanced testicular cancer] using cisplatin in 1978,” and has “investigated and published on cisplatin, as well as other platinum-based chemotherapies.” *Id.* ¶ 8.

Thus, we adopt Petitioner’s definition (*see* Pet. 7–8) as it is consistent with the prior art’s demonstration of the level of ordinary skill in the art at the time of the invention.⁵ In addition, we find that Dr. Earhart and Dr. Rassekh are both qualified to opine from the perspective of an ordinarily skilled artisan as both are persons of at least ordinary skill in the art. *See* Ex. 1007 ¶¶ 4–19, App. A; Ex. 2001 ¶¶ 9–35, App. A.

⁵ As explained below (*see infra* Section II.E), we agree with Petitioner that proposed substitute claim 2 is unpatentable under either party’s definition of skill level. *See* MTA Opp. 2; Revised MTA Opp. 2 n.1.

E. Proposed Amended Claim 2 Improperly Introduces New Matter

The '190 patent issued from application No. 16/112,195 (“the '195 application”), which is a continuation of application No. 15/826,243 (Ex. 2070, “the '243 application”). Ex. 1001, codes 21, 63.

Patent Owner points to disclosures in the '243 application to provide written description support for proposed substitute claim 2.⁶ *See* Revised MTA 3–10 (citing Ex. 2070). Specifically, Patent Owner asserts that the ototoxicity efficacy limitation is supported by the disclosures in the '243 application at page 13, pages 22–31 (Example 1), and Figures 3, 4, 9, 12, 23, 27, and 31. *Id.* at 4. Patent Owner asserts that the safety efficacy limitation is supported by the disclosures in the '243 application at pages 2, 4, 12, 20, claim 34, and Figures 5A-B, 13A-B, 16A-B, 17, 20A, 22A, 24A-B, 26A, 28A, 29A, 32A-B, and 35A. *Id.* at 5.

Petitioner disagrees. *See* Revised MTA Opp. 11–17. According to Petitioner, “Patent Owner has not demonstrated that the '243 application supports the ‘efficacy elements’ recited in revised Claim 2 for the specific

⁶ Our Rules also require that Patent Owner set forth the support in “the original disclosure of the patent for each claim that is added or amended.” 37 C.F.R. § 42.121(b)(1). In other words, Patent Owner in this case should have pointed out the support in the '195 application. *See* Ex. 1001, code 21. Patent Owner has not done so. Nonetheless, because the '243 application is a continuation of, and thus, shares substantially the same specification as the '195 application, we deem Patent Owner’s omission in identifying the support in the '195 application harmless. *See* Manual of Patent Examining Procedure § 201.07 (“The disclosure presented in the continuation must not include any subject matter which would constitute new matter if submitted as an amendment to the parent application.”).

patient population that involves a pediatric *medulloblastoma* patient that is *less than 5 years of age*, with a *localized and non-metastatic* tumor.” *Id.* at 11. We find Petitioner’s argument persuasive.

1. Case Law

As support for the ototoxicity limitation, Patent Owner relies on inferences drawn collectively from seven different tables and several passages in the ’243 application. *See* Revised MTA 4 (citing “Figs.3, 4, 9, 12, 23, 27, 31; p.13, ¶¶1,2; p.22–31 (Example 1”)); *see also id.* at 5 (relying on numerous other tables and passages as support for the safety efficacy limitation). Patent Owner cites *Nalpropion Pharms., Inc. v. Actavis Labs. FL., Inc.*, 934 F.3d 1344 (Fed. Cir. 2019) to support its argument that it can rely on “data from multiple tables in the specification to arrive at a claim limitation [to] satisf[y] the written description requirement.” Revised MTA 5. *Nalpropion* is distinguishable.

In *Nalpropion*, the Federal Circuit affirmed the district court’s finding that the challenged claim satisfied the written-description requirement. 934 F.3d at 1346. The challenged method claim recited administering a drug and requires that the drug have an *in vitro* dissolution profile as defined by recited ranges. *Id.* at 1347. The district court found that two tables in the specification adequately supported the dissolution data ranges. *Id.* at 1351. Specifically, it found the values in the two tables “fell squarely” within the claimed ranges. *Id.* at 1349. According to the Federal Circuit,

the [district] court was not convinced that relying on data from two tables presented a written description issue, noting that it found “nothing odd or invalidating about the inventors looking to different tables of dissolution data and other places in the

specification to determine the ranges for the claimed dissolution profile,” and finding that “multiple tests are necessarily required to establish a range.”

Id. at 1351 (citing *Orexigen Therapeutics, Inc. v. Actavis Labs. FL, Inc.*, 282 F. Supp. 3d 793, 803 (D. Del. 2017)). Notably, part of the court’s fact-finding involved evaluating and weighing experts’ testimony. *Id.*

The Federal Circuit stated that “these findings are supported by the record, and we do not disturb them.” *Id.* It explained that “[w]hile as a general matter written description may not be satisfied by so-called equivalent disclosure, in this case, buttressed by the district court’s fact-finding, and where the so-called equivalence relates only to resultant dissolution parameters rather than operative claim steps, we affirm the district court’s conclusion.” *Id.*

The facts in the instant case are different from those in *Nalpropion*. First, unlike the challenged claim there, which involved ranges, proposed amended claim 2 here does not recite any range. Thus, the district court’s reasoning that “multiple tests are necessarily required to establish a range” does not apply. *See Nalpropion*, 934 F.3d at 1351.

Second, unlike the two tables in *Nalpropion*, which discloses values that “fell squarely” within the claimed ranges (*id.* at 1349), none of the seven disparate figures and several passages Patent Owner relies on expressly discloses the ototoxicity efficacy limitation. *See infra* Section III.E.2.a. Instead, Patent Owner presents a series of complicated deductions, at least some of which are based on questionable premises (*see id.*), in order to calculate the reduction of ototoxicity in “**all potential scenarios.**” *See Revised MTA* 5–8.

Third, unlike the patentee in *Nalpropion*, who assisted the district court with expert testimony, Patent Owner here relies on, not expert testimony, but mere attorney argument to tie together the data from the disparate figures in the '243 application, depriving us of a sound evidentiary foundation on which to assess the accuracy of Patent Owner's representations and to piece together the many uncertain pieces of the puzzle. *See infra* Section III.E.2.a.

Similarly, we are not persuaded by Patent Owner's reliance on *Indivior UK Ltd. v. Reddy's Labs. S.A.*, 18 F.4th 1323, 1326 (Fed. Cir. 2021). *See* Reply 6. According to Patent Owner, in *Indivior*, the Federal Circuit upheld "a PTAB finding of written description support for claim 8 based on calculations attainable from tables which could be performed by a POSITA." *Id.* That may well be the case; but the facts here, again, are different, because there is no evidence to show the calculations performed by Patent Owner's counsel, without any expert testimony or other competent evidence to support, "could be performed by a POSITA." *See id.* Thus, *Indivior*, like *Nalpropion*, is distinguishable.

2. Ototoxicity Efficacy Limitation

a. *The '243 Application*

Patent Owner argues that the '243 application provides written-description support for the ototoxicity efficacy limitation. Revised MTA 4–8 (citing "Figs.3, 4, 9, 12, 23, 27, 31; p.13, ¶¶1,2; p.22–31 (Example 1)"). Even if we agree with Patent Owner that *Nalpropion* and *Indivior* permit Patent Owner to rely on the disparate disclosures in the '243 application for support, we still find that Patent Owner has not met

its burden to show that materials it points to provide sufficient support for the ototoxicity efficacy limitation.

We start with the text of the specification. Page 13 of the '243 application discloses “methods for reducing ototoxicity in patient[s] (i.e., pediatric patients) having a cancer and who are receiving a platinum based chemotherapeutic for treatment of the cancer.” Ex. 2070, 13. This passage, however, does not mention the claimed age (less than 5 years), cancer type (medulloblastoma), or extent of disease (localized and non-metastatic).

Similarly, Example 1 of the '243 application discloses “[p]articipants in the study were aged 1-18 years at study entry and were newly diagnosed with hepatoblastoma, germ cell tumour, medulloblastoma or CNS primitive neuroectodermal tumour, neuroblastoma, osteosarcoma, or other cancer types treated with cisplatin.” *Id.* at 22. Although Example 1 discloses the study results “[f]or participants younger than 5 years” (*id.* at 28), it does not specify the results for patients in that age group with the required cancer type (medulloblastoma) and the extent of disease (localized and non-metastatic), as recited in proposed substitute claim 2.

We next move on to the seven figures Patent Owner relies on. Among those figures, only Figures 4 and 9 show data on hearing loss in medulloblastoma patients. *Id.* at 60, 65. Neither of them, however, describes the age of the patients or the extent of disease. In fact, as explained below, considering information from Figures 23 and 31, it is clear the medulloblastoma patients in Figures 4 and 9 are *not* limited to those “less than 5 years of age.”

Figure 23 provides “the demographics of patients with medulloblastoma tumors enrolled in the study of Example 1.” *Id.* at 8. Figure 31 provides “the characteristics of the tumors for patients under the age of 5 enrolled in the study of Example 1.” *Id.* at 9. Both figures show that, for medulloblastoma patients, six in the observation arm were under five and eight in the STS arm were under five. *Id.* at 79; *see also* Ex. 1038, 202:20–203:1 (Dr. Rassekh testifying the same). Yet the hearing loss data in both Figures 4 and 9 are from ten medulloblastoma patients in the observation arm and nine in the STS arm. *Id.* at 60, 65. Thus, based on information in Figures 23 and 31, the data in Figures 4 and 9 could not have been entirely from patients “less than 5 years of age,” as proposed amended claim 2 requires.

Other figures Patent Owner relies on do not relate to hearing loss analysis. Figure 3 provides “the categorization of patient tumor types.” *Id.* at 6. It shows some patients have medulloblastoma and some have localized disease, but does not show patients “less than 5 years of age” with localized medulloblastoma, as proposed amended claim 2 recites. *Id.* at 59. Similarly, Figure 12 provides “the demographics of patients with localized tumors enrolled in the study of Example 1.” Ex. 2070, 7. It shows twelve medulloblastoma patients in the observation arm and nine in the STS arm. *Id.* at 68. Again, based on the information in Figures 23 and 31, the medulloblastoma patients with localized tumors in Figure 12 could not have been all patients “less than 5 years of age,” as proposed amended claim 2 requires.

The last figure Patent Owner relies on, Figure 27, “is a table providing the demographics of patients with osteosarcoma tumors enrolled in the study of Example 1.” Ex. 2070, 9. This figure does not appear to directly relate to the medulloblastoma patients. Yet Patent Owner manages to bring it into a theory that spans nearly three full pages, where Patent Owner sends us into a labyrinth, branched with if-this-then-that and either-or circumstances, and ending with a table of “**all potential scenarios**”—entirely without the guidance of expert testimony. *See* Revised MTA 5–8. What Patent Owner does not show is even a single actual scenario that provides support for the ototoxicity efficacy limitation, i.e., reduction of ototoxicity is at least about 60% for patients “less than 5 years of age” and with a “localized and non-metastatic medulloblastoma tumor.”

The lack of expert testimony supporting Patent Owner’s arguments is especially troublesome here. It is true that in some patent cases, “expert testimony will not be necessary because the technology will be easily understandable without the need for expert explanatory testimony.” *Centricut, LLC v. Esab Grp., Inc.*, 390 F.3d 1361, 1369 (Fed. Cir. 2004) (quotation marks omitted). This is not such a case. Regardless of the exact definition, the parties agree that the level of ordinary skill is high and may require input from a multidisciplinary team. *See supra* Section III.D. For example, the ’190 patent discloses the importance of statistical analysis in its exemplary studies. *See* Ex. 1001, 18:24–19:37, 32:46–33:3; *see also* Ex. 1038, 131:25–132:7 (Dr. Rassekh testifying that a statistician contributes to the development of study schema). This is because interpreting clinical study data is not always simple or straightforward.

One of Patent Owner's arguments illustrates this point well. Patent Owner argues that, in the '243 application, "Figs. 12 and 23 show that 3 medulloblastoma patients in the STS arm had prior cranial irradiation—2 had localized and 1 had disseminated disease." Revised MTA 6 (citing Ex. 2070, 68, 79). Although Figure 23 on its face shows that three medulloblastoma patients in the STS arm had prior cranial irradiation, and Figure 12 shows two patients with localized tumors had prior cranial irradiation, it is not apparent that the two patients referred to in Figure 12 have medulloblastoma, and not the other types of cancer listed therein.⁷ See Ex. 2070, 68, 79.

This is especially so in view of Figure 14, which provides "the characteristics of the tumors of patients with disseminated disease enrolled in the study of Example 1." *Id.* at 7. Figure 14 shows two patients with disseminated disease had prior cranial irradiation. *Id.* at 70. Thus, it is not clear to us, and Patent Owner does not explain, why among the three medulloblastoma patients in the STS arm who had prior cranial irradiation as shown in Figure 23, "2 had localized and 1 had disseminated disease," and not the other way around. See Revised MTA 6. Given the doubt casted by Figure 14, we accord Patent Owner's interpretation of the data little weight because it is unsupported by expert testimony or any other competent factual

⁷ Similarly, Patent Owner contends that "Fig. 12 also shows that 5 patients in the medulloblastoma OBS arm had prior cranial irradiation and localized disease." Revised MTA 6 (citing Ex. 2070, 68 (emphasis added)). Figure 12, however, does not show, and Patent Owner does not explain why, the five patients all have medulloblastoma, and not the other types of cancer listed therein.

evidence. *See In re Payne*, 606 F.2d 303, 315 (CCPA 1979) (“Arguments of counsel unsupported by competent factual evidence of record are entitled to little weight.”).

The Revised MTA is replete with such flaws. Another example brings the seemingly irrelevant Figure 27 into the picture. After making the unsubstantiated argument that, among the three medulloblastoma patients in the STS arm who had prior cranial irradiation as shown in Figure 23, “2 had localized and 1 had disseminated disease,” Patent Owner continues, again, without citing any evidence for support, that “[t]his accounts for 3 of the 4 patients >5 years of age.” Revised MTA 6. According to Patent Owner,

Fig. 27 shows the fourth patient in the STS arm who received prior cranial irradiation had osteosarcoma. [Ex. 2070,] 83. Depending on the hearing outcome of the osteosarcoma patient with prior cranial irradiation, either 1 (osteosarcoma patient had hearing loss) or 2 (osteosarcoma patient did not have hearing loss) of 3 medulloblastoma patients with prior cranial irradiation had hearing loss. *See Id.* at 28, ¶2; Ex. 2024, ¶102.

Revised MTA 7.

Paragraph 102 of Dr. Rassekh’s Declaration, however, does not support Patent Owner’s assertion. Instead, Dr. Rassekh testifies that

[t]he stratum of previous craniospinal radiation therapy contained 8 participants: 2 of the 4 (50%) participants receiving STS had hearing loss compared to 4 of the 4 (100%) participants not receiving STS, and none of the hearing loss assessments were changed at the 1 year follow up period. (Ex. 1001 at 20: 28-32, 47-50.)

Ex. 2024 ¶ 102. Dr. Rassekh offers no opinion on how many osteosarcoma patients with prior cranial irradiation and how many medulloblastoma patients with prior cranial irradiation had hearing loss.

We acknowledge that during deposition, Dr. Rassekh testified that “[t]here is data . . . in the patent figures that allows a person of skill in the art to look through the tables and show that the 60 percent bar is met in medulloblastoma, under five, localized.” Ex. 1038, 206:1–4; *see also id.* at 206:9–11 (“[I]f you look at the data, worse case scenario is a 60 percent bar is reached, based on the data presented in the patent.”).⁸ But Dr. Rassekh does not explain which figures and what data support his conclusion. Under our Rules, “[e]xpert testimony that does not disclose the underlying facts or data on which the opinion is based is entitled to little or no weight.” 37 C.F.R. § 42.65(a). Thus, we accord little weight to Dr. Rassekh’s deposition testimony on this issue. *See Koito Mfg. Co., Ltd. v. Turn-Key-Tech, LLC*, 381 F.3d 1142, 1152 (Fed. Cir. 2004) (stating that “[g]eneral and conclusory testimony” by an expert “does not suffice as substantial evidence of invalidity”).

In sum, we find the disclosures of the ’243 application that Patent Owner relies on do not provide sufficient written-description support for the ototoxicity efficacy limitation.

b. Multiple Dependent Claims

At the hearing, counsel for Patent Owner argued that original claims 8, 13, 14, and 21 of the ’243 application provide support for the ototoxicity efficacy limitation. *See* Tr. 20:14–21:6. According to counsel, claim 21 is multiple dependent that discusses reduction of ototoxicity in patients less than five. Claim 14,

⁸ We note that, this opinion is conspicuously absent in Dr. Rassekh’s Declaration.

medulloblastoma is included in the group of tumors that can be selected and then claim 13 is localized. These are all multiple dependents that go back through all of these and then claim 8 talks about at least 50 percent reduction in ototoxicity and that provides the guidance in the specification and if you look at page 13, paragraph 2 of Exhibit 2070, which is the application is filed, there's your at least 60 percent.

Id. at 20:24–21:6.

As a preliminary matter, in its briefing, including the Revised MTA and the Reply, Patent Owner does not rely on these claims of the '243 application for support of the ototoxicity efficacy limitation. Thus, this argument, presented only during oral argument, is waived. *See, e.g.*, Paper 12, 8 (“Patent Owner is cautioned that any arguments not raised in the response may be deemed waived.”).

Counsel for Patent Owner pointed to the safety efficacy support table, which lists claim 34. Tr. 21:13–20; *see also* Revised MTA 5 (citing Ex. 2070, claim 34 as support for the safety efficacy limitation). But even if we were to deem this reference to claim 34 in the discussion of the safety efficacy limitation as part of the discussion of the ototoxicity efficacy limitation, we still would reject Patent Owner's argument.

Claim 34 is directed to the “method of claims 1–33.” Ex. 2070, 53. Each of claims 9, 10, and 12–33 is a multiple dependent claim. *Id.* at 50–53. Claim 9 is directed to the “method of claims 1–8;” claim 10 is directed to the “method of claims 1–9;” claim 12 is directed to the “method of claims 1–11;” so on and so forth, and finally, claim 33 is directed to the “method of claims 1–32.” *Id.* Thus, claim 34 would encompass an astronomical number of methods, and Patent Owner does not point to blaze

marks directing us to the subject matter of the specific combination it attempts to rely on, i.e., the combination of claims 8, 13, 14, and 21. *See Fujikawa v. Wattanasin*, 93 F.3d 1559, 1570–71 (Fed. Cir. 1996).

And even if we were to arrive at the specific combination, the same problem still exists. Claim 13 recites “[t]he method of claims 1–12, wherein the cancer is localized.” Claim 14 recites “[t]he method of claims 1–13, wherein the cancer is selected from a germ cell tumor, hepatoblastoma, medulloblastoma, neuroblastoma, and osteosarcoma.” Claim 21 recites “[t]he method of claims 1–20, wherein the pediatric patient is about 5 years of age or less.”

To find support for the ototoxicity efficacy limitation in patient “less than 5 years of age” and with “a localized and non-metastatic medulloblastoma tumor,” an ordinarily skilled artisan would have to follow a “maze-like path, each step providing multiple alternative paths.” *Regents of the Univ. of Minnesota v. Gilead Scis., Inc.*, 61 F.4th 1350, 1357 (Fed. Cir. 2023). That, the Federal Circuit explains, “is not a written description of what might have been described if each of the optional steps had been set forth as the only option.” *Id.* Thus, we find the multiple dependent claims Patent Owner referred to during the oral argument do not provide sufficient written-description support for the ototoxicity efficacy limitation.

c. Summary

In sum, on this record, we agree with Petitioner that Patent Owner’s arguments amount to nothing but “a convoluted and disjointed story—with no supporting expert testimony.” *See* Revised MTA Opp. 13. We find Patent Owner has failed to meet its burden to show sufficient written-description

support for the ototoxicity efficacy limitation, i.e., reduction of ototoxicity of at least about 60% for patients “less than 5 years of age” and with a “localized and non-metastatic medulloblastoma tumor.” On this basis alone, we deny Patent Owner’s Revised MTA.

3. Overall Survival Efficacy Element

Patent Owner argues that the ’243 application provides written-description support for the overall survival efficacy limitation. Revised MTA 5 (citing “p.2, ¶¶1,2;p.4, ¶1; p.12, ¶2; p.20, ¶1; claim 34; Figs.5A-B, 13A-B, 16A-B, 17, 20A, 22A, 24A-B, 26A, 28A, 29A, 32A-B, 35A”). We are not persuaded.

The passages on pages 2, 4, 12, and 20 that Patent Owner relies on are general descriptions, and thus, do not provide sufficient support for the overall survival efficacy limitation. Figures 5A, 13A, 16A, 17, 20A, 22A, 24A, 26A, 28A, 29A, 32A, and 35A allegedly provide support for “relapse free survival,” a limitation independent from the overall survival limitation. *See* Revised MTA 8–9. Thus, we focus our discussion on, as Patent Owner does, Figures 5B, 13B, 16B, 24B, and 32B.

Figures 5B, 13B, 16B, 24B, and 32B show overall survival data for the observation arm and the STS arm. Ex. 2070, 61, 69, 72, 80, 88. As Patent Owner acknowledges, Figure 5B shows data for “the entire patient population for all cancers, all ages, and all extent of disease.” Revised MTA 9. Figure 13B shows data for localized disease and Figure 16B shows data for disseminated disease. *Id.* Neither specifies patient age or cancer type. Figure 24B shows data for medulloblastoma. *Id.* It does not specify patient age or extent of disease. Figure 32B shows data for patients <5 years

of age. *Id.* It does not specify cancer type or extent of disease. In other words, none of the figures Patent Owner relies on show data for patients “less than 5 years of age” and with a “localized and non-metastatic medulloblastoma tumor,” as proposed amended claim 2 requires. *See* Ex. 2070, 61, 69, 72, 80, 88.

Without citing any support, Patent Owner argues that “[w]hen looking at 2 independent events, a POSITA would know that to arrive at the likelihood that both events would occur (in this case a death of a patient with localized disease <5 years of age) is to multiply the individual probabilities.” Revised MTA 9–10. Again, without citing any support, Patent Owner argues that for medulloblastoma patients, “in the OBS arm, a 12% probability exists that the death was a patient <5 with localized disease,” and in the STS arm, that number is 17%. *Id.* at 10. That means, Patent Owner continues, without citing any support, “likely 0 and at most 1 death” in each arm. *Id.* Based on these unsupported assumptions and calculations, Patent Owner asserts that “a POSITA would conclude that there is not a significant difference in OS [overall survival] for localized medulloblastoma patients <5 years of age receiving STS.” *Id.*

As stated above, interpreting data from clinical study is often complicated. Without any expert testimony—or for that matter, any evidence—in support, we accord this mere attorney argument no weight. *See Payne*, 606 F.2d at 315. Thus, we find Patent Owner has not met its burden to show that the ’243 application provides written-description

support for the overall survival efficacy limitation.⁹ On this additional basis, we deny Patent Owner’s Revised MTA.

4. Summary

In sum, on this record, we find Patent Owner has failed to meet its burden to show sufficient written-description support for the ototoxicity efficacy limitation or the overall survival efficacy limitation for patients “less than 5 years of age” and with a “localized and non-metastatic medulloblastoma tumor.”

F. Petitioner’s Asserted Unpatentability Grounds

In light of our determination above that Patent Owner has not shown sufficient written-description support for proposed substitute claim 2 in the ’243 application, we do not reach the grounds of unpatentability set forth in Petitioner’s Opposition. *See* Revised MTA Opp. 17–25 (alleging non-enablement and obviousness).

IV. CONCLUSION¹⁰

We grant Patent Owner’s request to cancel original claim 1 of the ’190 patent, and deny Patent Owner’s request to add proposed substitute

⁹ Patent Owner cites to claim 34 of the ’243 application in the table but does not discuss it in its analysis. *See* Revised MTA, 5, 9–10. Claim 34 is another multiple dependent claim; it recites “[t]he method of claims 1–33.” Ex. 2070, 53. Even if we consider this claim, for the same reason explained above (*see supra* Section III.E.2.b), we find it insufficient to support the overall survival efficacy limitation.

¹⁰ Should Patent Owner wish to pursue amendment of the challenged claim in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner’s attention to the April 2019 *Notice*

claim 2 to the '190 patent. The results of this Decision are summarized in the following two tables.

Claim	35 U.S.C. §	References	Claim Shown Unpatentable	Claim Not shown Unpatentable
1	103	Freyer Presentation, Adherex Press Release ¹¹		
1	103	Brock, SIOPEL 6 Poster, Adherex Press Release ¹²		
Overall Outcome				

Motion to Amend Outcome	Claim
Original Claims Cancelled by Amendment	1
Substitute Claims Proposed in the Amendment	2
Substitute Claims: Motion to Amend Granted	
Substitute Claims: Motion to Amend Denied	2
Substitute Claims: Not Reached	

Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See 37 C.F.R. § 42.8(a)(3), (b)(2).*

¹¹ We do not reach this challenge because Patent Owner cancelled claim 1 in the Non-Contingent Motion to Amend. *See supra* Section II.

¹² We do not reach this challenge because Patent Owner cancelled claim 1 in the Non-Contingent Motion to Amend. *See supra* Section II.

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Patent Owner's Non-Contingent Motion to Amend is granted as to the request to cancel original claim 1 of the '190 patent;

FURTHER ORDERED that Patent Owner's Non-Contingent Motion to Amend is denied as to the request to add proposed substitute claim 2 to the '190 patent; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2022-00123
Patent 10,596,190 B2

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